

[images/cover/main.jpg — you at the bench, or your best build]

Mikko De Torres

Mechatronics, robotics and AI engineer

Mechanical design CAD 3D printing Robotics Machine learning

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Engineer, and the person who explains it

I design mechanical systems and then build them. My work runs from CAD modelling and 3D printing through electronics, control code and testing, and into the documentation and video that explain it.

Seven years in industry as a maintenance and automation engineer, then seven teaching robotics, control systems and machine learning at Batangas State University. I am finishing an MS in Artificial Intelligence on motion planning for six-axis manipulators.

What I bring to a team

A working knowledge of the whole loop — design it, build it, wire it, program it, test it, then document it well enough that someone else can pick it up. Teaching made the last part second nature.

Mechanical design and CAD

SolidWorks, Onshape, AutoCAD. Designed the structure and folding mechanism of a DOST-funded solar power station, built and deployed as a working prototype.

Electronics and code

Arduino, ROS, C++, Python, MATLAB. Built a 3-DOF manipulator controlled through ROS and Arduino, with the kinematics solved in Python.

Teaching and documentation

Robotics 1 and 2, control systems, machine learning and data science, microprocessors, physical systems modelling, basic workshop and machining. I run a YouTube channel on the same subjects.

Education

BS Mechatronics Engineering, 2012. MS Artificial Intelligence, in progress. Both at Batangas State University.

[the deployed unit in the field]

Collapsible solar power station for farms

A portable, foldable solar power unit for farm use, funded by DOST-TAPI and built to a working prototype.

The problem

What farms could not do without this, and why existing solar units did not fit — weight, cost, transport, setup time?

What I built

The mechanical structure and the folding mechanism, modelled in SolidWorks. The unit carries photovoltaic panels and a battery storage unit, and collapses so one team can move, deploy and store it without special equipment.

How

SolidWorks for the structure and mechanism, then fabrication drawings. I also produced the brochure and the instruction materials the users received.

Results

The proof: where it was deployed and what it delivered. Output, weight, setup time, units built.

What I would do differently

What fought you in the mechanism, and what you would change in a second version.

Role	Tools	Context	Year
Mechanical design lead	SolidWorks, fabrication drawings	DOST-TAPI funded project	2023–24

[the printed, assembled arm]

6-DOF robot arm kit

A six-axis arm, 3D printed and assembled into a teaching platform my students learn kinematics on.

The problem

Students were learning kinematics from equations and simulations only. What did that miss, and why did a physical arm change it?

What I built

What you printed, fitted, adjusted and assembled, and what the finished arm can do.

How

Printer and material, CAD package, controller and drive electronics, and what you had to modify to make the parts fit.

Results

How many students, across how many terms. What can they do now that they could not before?

What I would do differently

Print failures, tolerance problems, joints that needed reprinting.

Role	Tools	Context	Year
Built and integrated	3D printing, CAD, microcontroller	Robotics teaching platform	<i>Add</i>

[the build, or an RViz view]

3-DOF articulated manipulator on ROS and Arduino

A three-axis arm where the mechanical build, the electronics and the control software all had to meet.

The problem

What this arm was for, and why ROS rather than driving the servos straight from the Arduino.

What I built

The manipulator itself, with real-time control written in C++ under ROS and the kinematics solved in Python for motion planning.

How

ROS for the control layer, Arduino for the hardware interface, C++ for the real-time loop, Python for the kinematics.

Results

Show it working: reachable workspace, positioning accuracy, or a task it completes end to end.

What I would do differently

The integration problems — timing, serial latency, backlash.

Role	Tools	Context	Year
Built and programmed	ROS, Arduino, C++, Python	Add	Add

[a sample inspection or result plot]

Weld inspection by machine vision

A vision system that grades weld quality on stainless steel from HDR camera images.

The problem

Who was inspecting these welds before, how, and what that missed.

What I built

A full classification pipeline in Python that takes HDR camera images of a weld and classifies the defects in it.

How

Transfer learning on a pre-trained VGG16 network for feature extraction, then retrained for weld defect classification. HDR imaging holds detail across the bright and dark parts of the weld.

Results

The numbers: accuracy, dataset size, defect classes, and how it compared with inspection by eye.

What I would do differently

Dataset limits, lighting problems, classes it confused.

Role	Tools	Context	Year
Full pipeline	Python, HDR imaging, VGG16	Add	Add

[the SCARA V3 simulation]

Manipulator simulations in MATLAB

Simulations of every major manipulator type, written for the robotics courses I teach.

The problem

What third-year students were struggling to picture, and why the existing material did not get them there.

What I built

Simulations of the Cartesian, cylindrical, spherical, articulated and SCARA manipulator types, plus a 6-DOF SCARA V3, each covering kinematics, dynamics and control.

How

MATLAB with Peter Corke's Robotics Toolbox.

Results

How many students and terms, and what it replaced.

What I would do differently

Where the simulations diverge from the real hardware.

Role	Tools	Context	Year
Wrote all simulations	MATLAB, Robotics Toolbox	Robotics 1 & 2 teaching material	<i>Add</i>

[a planned path figure]

Motion planning for a six-axis manipulator

MS thesis: planning arm motion with neural eikonal fields, so the path avoids obstacles without hitting singularities.

The problem

What existing planners do badly for a six-axis arm, and what singularities cost in practice.

What I built

What the method does and what you implemented yourself.

How

Your stack, the network, the training setup, how you evaluated it.

Results

Whatever you have so far, even partial — plots, success rates, comparisons against a baseline planner.

What I would do differently

In progress. Say so plainly and give the current state.

Role	Tools	Context	Year
Sole author	Add your stack	MS Artificial Intelligence thesis	In progress

[channel screenshot, or two or three thumbnails]

Teaching, filming and open material

The engineering is only half of it. I also make the material that explains it.

Mechatronics and Robotics Engineering on YouTube

I script, film and edit technical videos on robotics, data science in Excel, Git and GitHub, and basic workshop and machining.

Course repositories

Robotics_MEXE_3rdYearCourse — kinematics, control algorithms and simulation tasks for Robotics 1 and 2.

MexEE402_AI — data science, machine learning and model productionization.

Print documentation

Brochure and instruction materials for the solar power station, written for the farmers who would use it.

Role	Tools	Context	Year
Author and presenter	Video, GitHub, technical writing	Public teaching material	2019–

Let's build something.

If any of this is close to what you are working on, I would be glad to talk it through.

[email]

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[portfolio site]

[LinkedIn]

[YouTube]

[GitHub]